

Calculus

Մաթեմատիկական անալիզ

COURSE CODE	MTH-CA
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Typically Grade 11 or 12
DURATION	108–126 instructional hours (approx. 90 lessons)
PREREQUISITE	Precalculus (MTH-PC) or equivalent, with demonstrated algebraic and trigonometric fluency.
LEADS TO	AP Calculus AB (MTH-APCA) or BC (MTH-APCB) for students seeking college credit; university calculus otherwise

COURSE DESCRIPTION

Դասընթացի նկարագրություն

This is a complete first course in single-variable calculus for students who want the subject without the constraints of the AP examination — because their school does not offer it, because they want a slower and deeper treatment, or because they intend to take the AP course afterwards with the content already familiar. It covers limits and continuity, the derivative and its applications, the definite integral and the Fundamental Theorem, techniques and applications of integration, and an introduction to differential equations and series.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Calculus is two ideas — the derivative and the integral — connected by one theorem, and everything else is technique. Վարժարան teaches it that way. Each new concept is developed graphically, numerically, analytically, and verbally, and students are expected to move among these registers on demand, because a student who can only differentiate symbolically has learned a procedure rather than a subject. Teachers are also asked to hold the algebraic line: most errors in a calculus course are algebra errors, and diagnosing them as such — rather than reteaching the calculus — is what actually resolves them.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Manipulates rational and radical expressions without hesitation	The dominant source of error in calculus; remediate before Unit 2 rather than during it.
Produces exact trigonometric values and applies identities	Required continuously from the derivative rules onward.
Applies exponent and logarithm laws fluently	Required for exponential and logarithmic differentiation and integration.
Analyzes a function's domain, asymptotes, and transformations	Prerequisite for curve sketching and for limit analysis.
Evaluates limits informally and understands the difference quotient	If absent, Precalculus Unit 8 should be completed first.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Evaluate limits analytically, graphically, and numerically, including limits at infinity and one-sided limits.
- 2 Determine continuity and classify discontinuities, and apply the Intermediate and Extreme Value theorems.
- 3 Define the derivative as a limit and interpret it as a slope and as an instantaneous rate of change.
- 4 Differentiate using the power, product, quotient, and chain rules and the derivatives of the elementary functions.
- 5 Perform implicit differentiation and find derivatives of inverse functions.
- 6 Apply the derivative to related rates, linearization, and optimization problems.
- 7 Use the first and second derivatives to analyze and sketch the graph of a function.
- 8 State and apply the Mean Value Theorem and understand its role.
- 9 Understand the definite integral as a limit of Riemann sums and as accumulated change.
- 10 State and apply both parts of the Fundamental Theorem of Calculus.
- 11 Integrate using substitution, integration by parts, and partial fractions.
- 12 Apply integration to area, volume, arc length, average value, and physical accumulation problems.
- 13 Solve separable differential equations and interpret slope fields.
- 14 Analyze sequences and series for convergence and represent functions as power series.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 11-12 LESSONS

Limits and Continuity

Մահմաններ և անընդհատություն

TOPICS COVERED

- The limit concept, developed graphically and numerically
- One-sided limits and the existence of a limit
- Limit laws and their application
- Algebraic techniques: factoring, rationalizing, common denominators
- Indeterminate forms and what they indicate
- Limits involving trigonometric functions; the fundamental trigonometric limits
- The squeeze theorem
- Infinite limits and vertical asymptotes
- Limits at infinity and horizontal asymptotes
- End behaviour of rational, exponential, and logarithmic functions
- Continuity at a point and on an interval
- Classifying discontinuities and removable discontinuity
- The Intermediate Value Theorem and its applications
- The formal ε - δ definition of a limit, introduced for understanding rather than computation

LEARNING OBJECTIVES

- 1 Evaluate limits by an appropriately chosen analytic technique and justify the choice.
- 2 Determine continuity and classify any discontinuity with reasons.
- 3 Apply the Intermediate Value Theorem to establish the existence of a root.
- 4 Explain what the formal definition of a limit says, in words.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Evaluates limits at 90% accuracy across all techniques.
- Classifies discontinuities correctly in 9 of 10 trials.
- Applies the IVT with a correct hypothesis check in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A limit is assumed not to exist whenever the function is undefined at the point.	The limit describes approach, not arrival. A removable discontinuity has a limit and no function value.

MISCONCEPTION	RECOMMENDED RESPONSE
Indeterminate forms are treated as answers.	0/0 and ∞/∞ signal that further work is required. They are questions, not conclusions.
Limit laws are applied to indeterminate forms without checking applicability.	The laws require the individual limits to exist and to be finite. Check before applying.

TEACHING NOTE

The ε - δ definition is included for understanding, not for computation. One lesson devoted to what it says — and to why an intuitive definition is insufficient — repays itself for any student continuing to university mathematics.

UNIT 2 • 12-13 LESSONS**The Derivative**

Ածանցյալ

TOPICS COVERED

- Average and instantaneous rate of change
- The tangent line problem and the derivative as a limit
- The derivative at a point and the derivative function
- Differentiability and its relationship to continuity
- Where a function fails to be differentiable
- The graphical relationship between f and f'
- Power rule, constant multiple, sum and difference rules
- Derivatives of sine and cosine from the definition
- Product and quotient rules
- The chain rule and composite functions
- Derivatives of all six trigonometric functions
- Derivatives of exponential and logarithmic functions
- Logarithmic differentiation
- Implicit differentiation
- Derivatives of inverse functions and of inverse trigonometric functions
- Higher-order derivatives and their interpretations

LEARNING OBJECTIVES

- 1 Compute a derivative from the definition and explain what it measures.
- 2 Differentiate any elementary function using the appropriate rules in combination.
- 3 Perform implicit differentiation and find the equation of a tangent line to an implicit curve.
- 4 Sketch f' from a graph of f and vice versa.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Differentiates at 92% accuracy including nested chain rule applications.
- Performs implicit differentiation correctly in 9 of 10 trials.
- Sketches derivative graphs from function graphs in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The chain rule is omitted for a composite function.	Identify the outer and inner functions explicitly, in writing, before differentiating. This is the most common error in the entire course.
The product rule is applied as the product of the derivatives.	Verify with a simple example where both answers can be computed independently.
In implicit differentiation, dy/dx is omitted when differentiating a y term.	y is a function of x , so the chain rule applies. Write $y(x)$ explicitly for the first several examples.

TEACHING NOTE

Deriving the sine and cosine derivatives from the definition, using the fundamental trigonometric limits, is worth the time. It also justifies why radians are required — a point that otherwise remains an unexplained rule.

UNIT 3 • 13-14 LESSONS

Applications of the Derivative

Ածանցյալի կիրառություններ

TOPICS COVERED

- Related rates: setting up and solving
- Linearization and differentials; error estimation
- Extreme values, critical points, and the Extreme Value Theorem
- Rolle's theorem and the Mean Value Theorem
- The first derivative test and intervals of increase and decrease
- Concavity, inflection points, and the second derivative test
- Complete curve sketching from f , f' , and f''
- Optimization: modelling, constraint, and verification
- Optimization problems in geometry, business, and physics
- Motion on a line: position, velocity, acceleration, speed
- Interpreting the sign of velocity and acceleration
- L'Hôpital's rule and indeterminate forms
- Newton's method for approximating roots

LEARNING OBJECTIVES

- 1 Solve a related rates problem with a labelled diagram and a stated relationship.
- 2 Find and classify all extrema and inflection points and sketch a complete curve.
- 3 Solve an optimization problem, including verification that the extremum is of the required type.
- 4 Interpret motion on a line from the signs of velocity and acceleration.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves related rates problems in 4 of 5 tasks with correct setup.

- Produces complete curve sketches with all features justified in 4 of 5 tasks.
- Solves optimization problems including verification in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
In related rates, values are substituted before differentiating.	Differentiate the general relationship first, then substitute the instantaneous values. Substituting early destroys the variables that are changing.
An optimization answer is reported without verifying it is a maximum rather than a minimum.	Use the first or second derivative test, or check the endpoints. Verification is part of the answer.
A sign change of f'' is assumed at every zero of f''.	An inflection point requires an actual change in concavity. Test on both sides.

TEACHING NOTE

Optimization is where calculus becomes visibly useful, and it is also where modelling ability matters more than technique. Require a diagram, a stated objective function, a stated constraint, and a stated domain before any differentiation.

UNIT 4 • 11-12 LESSONS

The Definite Integral

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TOPICS COVERED

- The area problem and approximation by rectangles
- Riemann sums: left, right, midpoint, and trapezoidal
- The definite integral as a limit of Riemann sums
- Properties of the definite integral
- The definite integral as accumulated change
- Interpreting an integral in context with units
- Antiderivatives and the indefinite integral
- The Fundamental Theorem of Calculus, Part 1
- The Fundamental Theorem of Calculus, Part 2
- Why the two parts together connect the derivative and the integral
- Functions defined by integrals and their derivatives
- The average value of a function and the Mean Value Theorem for integrals
- Integration by substitution
- Definite integrals with substitution and changing limits
- Numerical integration and error considerations

LEARNING OBJECTIVES

- 1 Approximate a definite integral by several Riemann sums and state whether each over- or underestimates.

- 2 State both parts of the Fundamental Theorem precisely and explain what each says.
- 3 Differentiate a function defined by an integral, including with a variable upper limit and the chain rule.
- 4 Interpret a definite integral in an applied context with correct units.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes Riemann sums and identifies over/underestimation in 9 of 10 trials.
- Applies both parts of the Fundamental Theorem correctly at 90% accuracy.
- Interprets integrals in context with units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The limits of integration are not changed when substituting in a definite integral.	Either change the limits or convert back to the original variable. State which approach is being used, in writing.
A definite integral is assumed to give area even when the function is negative.	It gives signed area. For total area, integrate the absolute value or split at the zeros.
The Fundamental Theorem is applied to a function that is not continuous on the interval.	Check the hypotheses. Discontinuities within the interval require the integral to be split.

TEACHING NOTE

The interpretation of an integral in context with units is where students most often falter and where applied problems live. Ask 'the integral of what, with respect to what, gives what?' for every applied integral in the unit.

UNIT 5 • 13-14 LESSONS

Techniques and Applications of Integration

Ինտեգրման եղանակներ և կիրառություններ

TOPICS COVERED

- Integration by parts and the tabular method
- Trigonometric integrals and trigonometric substitution
- Integration by partial fractions
- Improper integrals and tests for convergence
- Area between two curves, integrating with respect to x or y
- Volumes by the disk and washer methods
- Volumes by cylindrical shells
- Volumes by known cross-sections
- Arc length and surface area of revolution
- Average value of a function in applied contexts
- Accumulation problems: total distance, total cost, total change
- Work, fluid force, and centre of mass
- Choosing an integration technique systematically

LEARNING OBJECTIVES

- 1 Choose an appropriate integration technique and justify the choice before computing.
- 2 Set up and evaluate a volume integral by the most convenient method.
- 3 Determine whether an improper integral converges and evaluate it if so.
- 4 Model and solve an applied accumulation problem with correct units.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Selects and executes the correct technique in 9 of 10 integrals.
- Sets up volume integrals correctly by disk, washer, and shell in 4 of 5 tasks.
- Solves applied accumulation problems with correct units in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The washer method is applied with the radii subtracted before squaring.	The difference of squares, not the square of the difference. Write both radii separately before assembling the integrand.
The wrong variable of integration is chosen, making the setup far harder than necessary.	Sketch the region and the representative slice first. The slice orientation determines the variable.
An improper integral is evaluated without a limit.	Improper integrals are defined as limits. Write the limit explicitly, every time.

TEACHING NOTE

Setting up the integral is the assessed skill in this unit; evaluation is secondary. Require a sketch and a labelled representative slice for every volume problem before any integration begins.

UNIT 6 · 13-14 LESSONS**Differential Equations and Series**

Դիֆերենցիալ հավասարումներ և շարքեր

TOPICS COVERED

- Differential equations and their solutions
- Slope fields and reading solution behaviour from them
- Separable differential equations
- Exponential growth and decay as a differential equation
- The logistic differential equation and its solution behaviour
- Euler's method for numerical approximation
- Sequences and their convergence
- Infinite series and partial sums
- Geometric series and the n th term test
- Integral, comparison, limit comparison, ratio, and root tests
- Alternating series and the alternating series error bound
- Absolute and conditional convergence
- Power series, radius and interval of convergence
- Taylor and Maclaurin series and the Lagrange error bound
- Series for the elementary functions and their manipulation

LEARNING OBJECTIVES

- 1 Solve a separable differential equation with an initial condition and interpret the solution.
- 2 Sketch and interpret a slope field and match it to a differential equation.
- 3 Determine convergence of a series using an appropriately chosen and justified test.
- 4 Construct a Taylor series and bound the error of a truncation.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves separable differential equations at 90% accuracy.
- Chooses and correctly applies a convergence test in 9 of 10 series.
- Constructs Taylor series and applies error bounds in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The constant of integration is omitted or applied before separating.	Include it and apply the initial condition afterwards. The constant is where the initial condition enters.
A convergence test is applied without checking its hypotheses.	Each test has conditions. State the test and verify its conditions before using it.

MISCONCEPTION	RECOMMENDED RESPONSE
The nth term test is used to conclude convergence.	It can only establish divergence. If the terms tend to zero, the test is inconclusive and another test is needed.

TEACHING NOTE

Series is the most abstract material in the course and benefits from constant numerical grounding: compute partial sums, graph them, and observe the behaviour. Students who only manipulate tests symbolically rarely develop any sense for what convergence means.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Precalculus algebra and trigonometry, weighted heavily toward rational expressions and exact values.
Unit Check	End of each unit · 60 min	Unit mastery criteria with required calculator-free and calculator-permitted sections.
Algebra Fluency Probe	Every 2nd lesson · 5 min	The algebraic manipulations calculus assumes; continues all year, since decay here is the primary cause of mid-year decline.
Conceptual Check	Once per unit	Questions requiring verbal or graphical reasoning with no computation permitted — the check on whether the student understands or only executes.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; weighted toward the derivative and its applications.
End-of-Course Assessment	After Unit 6	All course outcomes in a format resembling a university first-year final examination.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard Year	2 lessons/week, 75 min	Full course in approximately 45 weeks.
Accelerated	3 lessons/week	Full course in approximately 30 weeks.
AP Preparation	2–3 lessons/week	Full course followed by targeted AP examination preparation; see MTH-APCA and MTH-APCB.
Support / Co-Seated	2 lessons/week alongside school	Paced to the student's school or university syllabus.
Summer Program	4 lessons/week, 8 weeks	Units 1–3 — limits and the derivative — for students entering Calculus in the autumn.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Seventy-five minutes per lesson: fifteen to twenty problems in three parts — current content, cumulative review including algebra maintenance, and one extended problem requiring a full written solution with justification. Applied problems require a diagram, stated variables, and an interpretive sentence with units. Non-calculator work appears in every assignment. Students maintain an error log distinguishing algebra errors from calculus errors, which typically reveals that the great majority are the former — a diagnosis that changes how the student practices.

PROGRESS MILESTONES REPORTED TO PARENTS

Ուսումնական նվաճումներ

These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Evaluates limits analytically, graphically, and numerically.
- ◆ Determines continuity and classifies discontinuities.
- ◆ Computes derivatives from the definition and by all standard rules.
- ◆ Performs implicit and logarithmic differentiation.
- ◆ Sketches f' from f and reasons across representations.
- ◆ Solves related rates and optimization problems with full modelling.
- ◆ Analyzes and sketches curves using the first and second derivatives.
- ◆ Understands the definite integral as a limit of Riemann sums and as accumulated change.
- ◆ States and applies both parts of the Fundamental Theorem of Calculus.
- ◆ Integrates by substitution, parts, trigonometric methods, and partial fractions.
- ◆ Computes areas, volumes, arc lengths, and average values.
- ◆ Solves separable differential equations and interprets slope fields.

- ◆ Determines series convergence with justified tests and constructs Taylor series.
- ◆ Completed Calculus — ready for AP examinations or university mathematics.

MATERIALS AND RESOURCES

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- Graphing calculator (TI-84 family or equivalent) with a stated policy on permitted use.
- Computer algebra system for verification, used only after work is completed by hand.
- Dynamic graphing software for slope fields, Riemann sums, and solids of revolution.
- Վարժարան Calculus problem bank — approximately 2,600 items including 500 calculator-free items and 200 conceptual (non-computational) items.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.